Samuel Horner

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EDUCATION

Computer Engineering Technology (Computing Science), Diploma 2024 – Present
Algonquin College, Ottawa, ON
Relevant coursework: Compilers, Database Design (SQL, ERDs), Object-Oriented Programming, Socket Programming

PROJECTS

Git-like Version Control System | C++17, Qt, CMake github.com/Sam-Horner-42/CPPMiniVersionControl
- Co-developed a local Git-style version control system with a team of 5 (147 commits): commits, line-by-line diffs, file hashing and one-click revision restore.
- As Software Architect and UI Developer, designed three decoupled layers (Qt GUI, core engine, storage) and built the full Qt GUI: repository view, commit log, diff inspector and staging.
- Built a file-watching system that detects modified files so the repository can update; ran 100+ tests to verify versioning behaviour.

Stelline Compiler (front-end) | C github.com/Sam-Horner-42/A1Compilers
- Built a compiler front-end for the Stelline teaching language solo, with no external libraries or parser generators: growable input buffer, table-driven finite-state lexer and recursive-descent parser for an EBNF grammar.
- Implemented three buffer growth modes (fixed, additive, multiplicative with amortised O(1) appends) and error reports with exact line and character position; builds on GCC, Clang and MSVC.

Blasteroids | Lua, PICO-8 skwidge.itch.io/blasteroids
- Designed and shipped a browser-playable endless shoot 'em up solo over 8 months within PICO-8's 128×128 screen, 16-colour palette and code-size cap; created all pixel art, music and sound effects.
- Designed three weapons (blaster, rockets, laser) and a shield mechanic to rebalance difficulty; used AI assistance for part of the Lua gameplay code, then integrated, tested and tuned it within console limits.

Cookie Conquest | Java fabe-da-babe.itch.io/cookie-conquest
- Co-developed a dice-pool strategy roguelike with a team of 3 for Brackeys Game Jam 2025.2, writing the majority of the game logic; shipped a single-download Windows build with a bundled Java runtime.

EXPERIENCE

Front-line Technology Support Volunteer, Connected Canadians, Ottawa, ON Dec 2025 – Present
- Provide one-to-one technical support and education to older adults navigating digital devices.
- Diagnose hardware and software usability issues and write clear instructions that explain technical concepts to non-technical users.

Kitchen Volunteer, The Ottawa Mission, Ottawa, ON Sep 2021 – Present
- Supported community meal preparation, helping provide over 100 meals per shift.

Concierge, Spiritleaf Inc., Ottawa, ON Feb 2023 – Oct 2026
- Served a wide range of customers daily; tracked sales and stock levels in point-of-sale and inventory management software.
- Followed regulatory compliance standards and contributed to internal processes for tracking and managing products.

Counter Staff, Tim Hortons, Ottawa, ON Apr 2022 – Feb 2023
- Handled fast-paced, high-volume service (100+ customers per shift) while maintaining regulatory standards.

Floor, Garden Centre & Lumberyard Staff, BMR, Winchester, ON Jul 2015 – Aug 2021
- Updated stock levels in inventory tracking software and prioritized daily tasks based on customer demand.

TECHNICAL SKILLS

Languages: C, C++, Java, C#, Python, Lua, PHP, SQL (MySQL), Assembly, HTML/CSS
Frameworks, Libraries & Engines: Qt, Bootstrap, REST APIs, Godot, Love2D, PICO-8
Tools: Git, GitHub, CMake, VS Code, Visual Studio, IntelliJ, Eclipse, Wireshark, VMware, Multisim, UML
Systems & Networking: Linux (CLI scripting), Windows 11, PowerShell, batch files, socket programming, network troubleshooting
Practices: Object-oriented programming, relational modeling (ERDs), software documentation, Agile/team collaboration

CERTIFICATIONS

Accessibility for Ontarians with Disabilities Act (Sep 2021) | Standard CPR C & AED Training (Jan 2024)